

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

June 2023 Reconstructed Past Paper

DETAILED SOLUTIONS

Original question followed by a full worked solution

29 adaptive question sections 97 marks

Selected from Linear Higher Papers 1H and 2H

Dr Eslam Ahmed

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Assistant Lecturer, Faculty of Engineering, Cairo University

B.Sc. Electrical Engineering (2017), Distinction with Honours

M.Sc. Mathematics (2023)

PhD Candidate

eliteigcse.com • WhatsApp +20 112 000 9622

About this reconstructed paper

Each question section appears in the same order as the Student book: the complete source demand is followed by its demand-local worked solution and final answer.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 1; the selection contains 97 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

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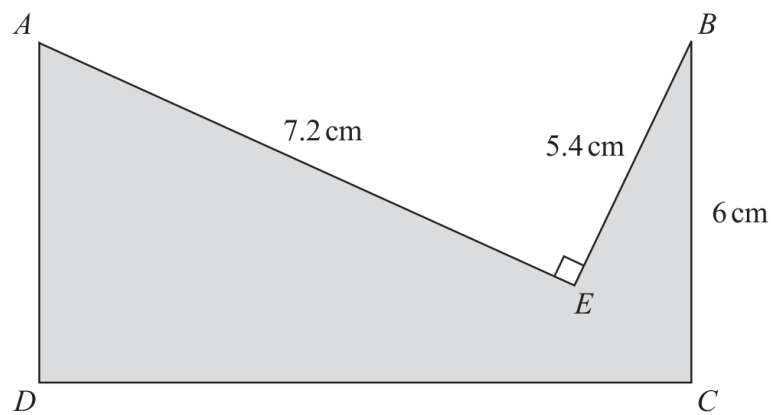
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Question 01

4MA1-1H Q5

ORIGINAL QUESTION**4 marks**

- 5 The diagram shows a shaded shape $AEBCD$ made by removing triangle AEB from rectangle $ABCD$



$$AE = 7.2 \text{ cm} \quad BE = 5.4 \text{ cm} \quad BC = 6 \text{ cm} \quad \text{angle } AEB = 90^\circ$$

Work out the perimeter of the shaded shape.

..... cm

Worked Solution 01

Perimeter with Pythagoras

MODULAR UNIT 1

4 marks

Find the missing rectangle side

$$AB = \sqrt{7.2^2 + 5.4^2} = 9.$$

Add the shaded boundary

$$7.2 + 5.4 + 6 + 9 + 6 = 33.6.$$

Final answer

33.6 cm

Question 02

4MA1-1H Q6

ORIGINAL QUESTION

2 marks

6 (a) Simplify $(2c^4d^7)^3$

.....
(2)

Worked Solution 02

Power of a monomial

MODULAR UNIT 1

2 marks

Apply the outer power

$$(2c^4d^7)^3 = 2^3c^{12}d^{21}.$$

Final answer

$$8c^{12}d^{21}$$

Question 03

4MA1-1H Q6

ORIGINAL QUESTION

1 marks

(b) Find the value of $5y^0$ where $y > 0$

(1)

Worked Solution 03

Zero index

MODULAR UNIT 1

1 marks

Use the zero-power rule

$$y^0 = 1 \quad (y > 0), \quad 5y^0 = 5.$$

Final answer

5

Question 04

4MA1-1H Q6

ORIGINAL QUESTION

2 marks

(c) Factorise fully $16a^2b^3 + 20a^3b$

.....
(2)

Worked Solution 04

Common-factor factorisation

MODULAR UNIT 1

2 marks

Take the greatest common factor

$$16a^2b^3 + 20a^3b = 4a^2b(4b^2 + 5a).$$

Final answer

$$4a^2b(4b^2 + 5a)$$

Question 05

4MA1-1H Q6

ORIGINAL QUESTION

3 marks

(d) (i) Factorise $x^2 + 9x - 22$

.....
(2)

(ii) Hence solve $x^2 + 9x - 22 = 0$

.....
(1)

Worked Solution 05

Factorise then solve

MODULAR UNIT 1

3 marks

Factorise

$$x^2 + 9x - 22 = (x + 11)(x - 2).$$

Use the zero-product rule

$$x + 11 = 0 \text{ or } x - 2 = 0.$$

Final answer

$$x = -11 \text{ or } x = 2$$

Question 06

4MA1-1H Q13

ORIGINAL QUESTION

3 marks

13 A is the point with coordinates $(-5, 12)$

B is the point with coordinates $(19, -48)$

Find an equation of the straight line that passes through the points A and B

Worked Solution 06

Equation of a line through two points

MODULAR UNIT 1**3 marks****Find the gradient**

$$m = \frac{-48 - 12}{19 - (-5)} = \frac{-60}{24} = -2.5.$$

Find the intercept

$$12 = -2.5(-5) + c \Rightarrow c = -0.5.$$

Final answer

$$y = -2.5x - 0.5$$

Question 07

4MA1-1H Q14

ORIGINAL QUESTION

3 marks

14 Factorise fully $50g^2 - 18$

Worked Solution 07

Difference of two squares

MODULAR UNIT 1

3 marks

Take out 2

$$50g^2 - 18 = 2(25g^2 - 9).$$

Factor the difference

$$25g^2 - 9 = (5g + 3)(5g - 3).$$

Final answer

$$2(5g + 3)(5g - 3)$$

Question 08

4MA1-1H Q15

ORIGINAL QUESTION

3 marks

15 (a) $\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n
Show your working clearly.

$n = \dots\dots\dots$
(3)

Worked Solution 08

Numerical indices

MODULAR UNIT 1

3 marks

Write every term as a power of 2

$$\sqrt{2} = 2^{1/2}, \quad 8^3 = 2^9, \quad 16^{3/2} = 2^6.$$

Subtract indices

$$2^{1/2} \div 2^{9-6} = 2^{1/2-3} = 2^{-5/2}.$$

Final answer

$$n = -\frac{5}{2} = -2.5$$

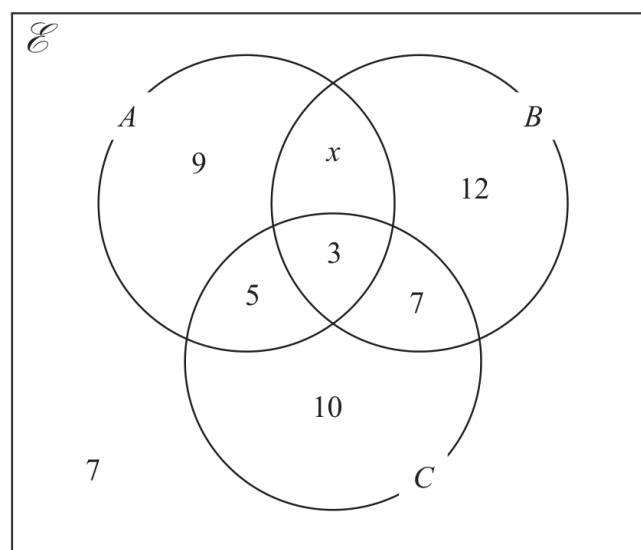
Question 09

4MA1-1H Q16

ORIGINAL QUESTION**3 marks**

16 The Venn diagram shows a universal set $\mathcal{E}$ and sets A , B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x = \dots\dots\dots$
(1)

(b) Find $n(A')$

$\dots\dots\dots$
(1)

(c) Find $n(B' \cap C)$

$\dots\dots\dots$
(1)

Worked Solution 09

Venn-diagram counts

MODULAR UNIT 1

3 marks

Use the union total

$$9 + 12 + x + 5 + 7 + 3 = 42 \Rightarrow x = 6.$$

Count outside A

$$n(A') = 12 + 7 + 10 + 7 = 36.$$

Count C but not B

$$n(B' \cap C) = 5 + 10 = 15.$$

Final answer

$$x = 6; \quad n(A') = 36; \quad n(B' \cap C) = 15$$

Question 10

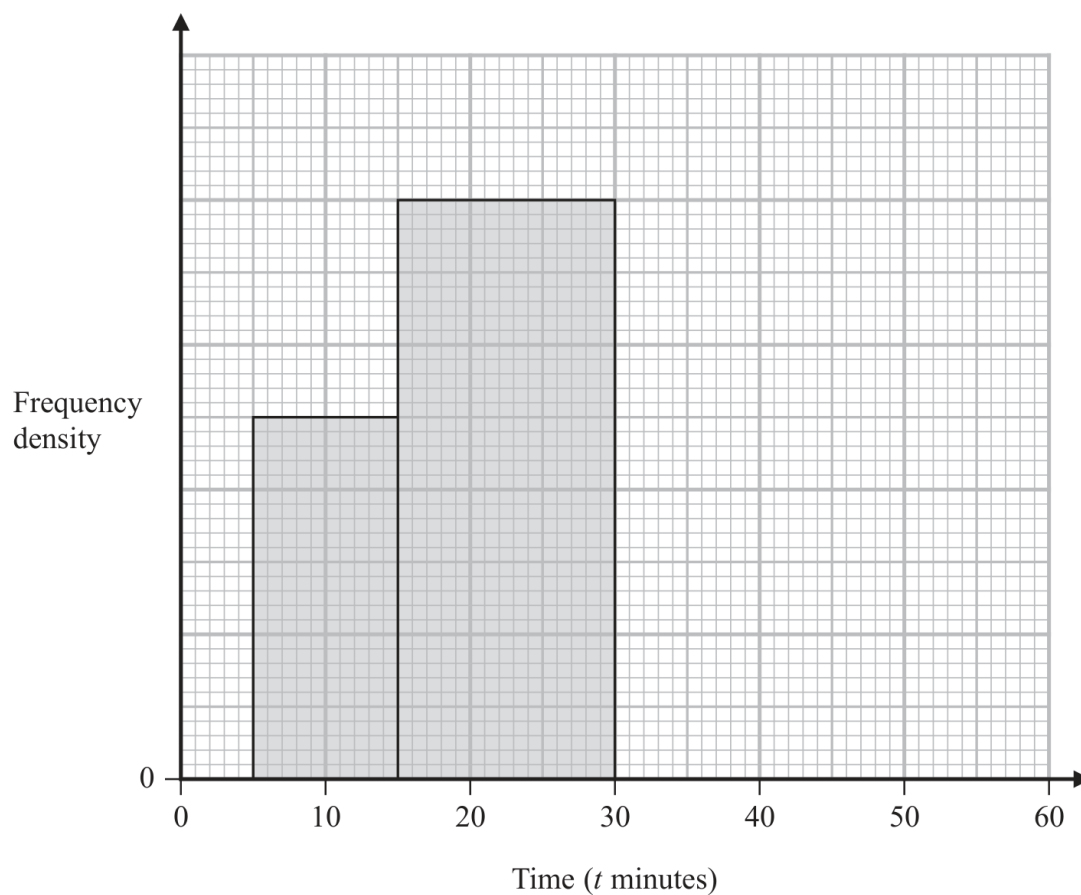
4MA1-1H Q18

ORIGINAL QUESTION

4 marks

- 18 The incomplete table and incomplete histogram give information about the times, in minutes, that 140 people waited at a station for a train.

Time (t minutes)	Frequency
$0 < t \leq 5$	23
$5 < t \leq 15$	
$15 < t \leq 30$	
$30 < t \leq 40$	18
$40 < t \leq 60$	14



Complete the table and the histogram.

Worked Solution 10

Complete a grouped table and histogram

MODULAR UNIT 1

4 marks

Find the two missing frequencies

$$140 - (23 + 18 + 14) = 85, \quad 85 \text{ is shared in the ratio } 10(2.5) : 15(4) = 25 : 60.$$

Complete the table

23, 25, 60, 18, 14.

Use frequency density

$$\frac{23}{5} = 4.6, \quad \frac{25}{10} = 2.5, \quad \frac{60}{15} = 4, \quad \frac{18}{10} = 1.8, \quad \frac{14}{20} = 0.7.$$

Final answer

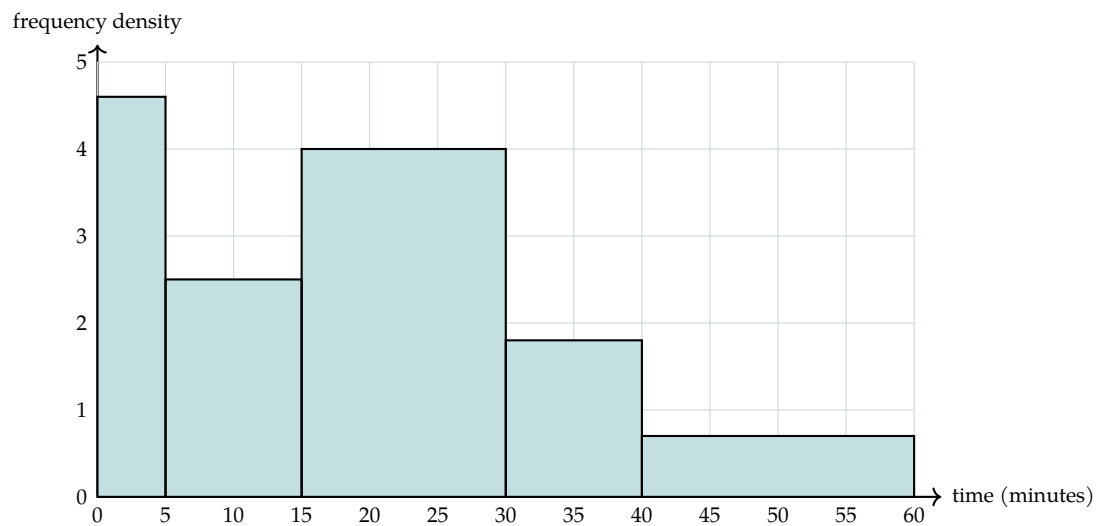
Frequencies 23, 25, 60, 18, 14; densities 4.6, 2.5, 4, 1.8, 0.7

Completed diagram 10

Complete a grouped table and histogram

MODULAR UNIT 1

4 marks

Completed solution diagram: Completed histogram

Question 11

4MA1-1H Q22

ORIGINAL QUESTION

6 marks

- 22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

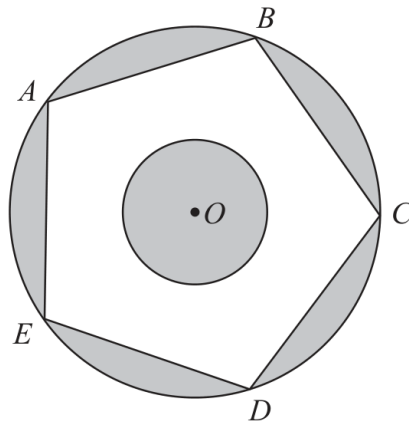


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Worked Solution 11

Equal shaded and unshaded areas

MODULAR UNIT 1

6 marks

Find the circumradius

$$R = \frac{4}{\sin 36^\circ} = 6.805 \dots$$

Find the two main areas

$$A_{\text{circle}} = \pi R^2 = 145.489 \dots, \quad A_{\text{pentagon}} = 5 \left(\frac{1}{2} R^2 \sin 72^\circ \right) = 110.111 \dots$$

Use the equal-area condition

$$A_{\text{circle}} - A_{\text{pentagon}} + \pi r^2 = A_{\text{pentagon}} - \pi r^2.$$

Solve

$$r = \sqrt{\frac{2A_{\text{pentagon}} - A_{\text{circle}}}{2\pi}} = 3.44 \dots$$

Final answer

3.45 cm

Question 12

4MA1-2H Q1

ORIGINAL QUESTION

3 marks

1 Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

Worked Solution 12

Divide mixed numbers

MODULAR UNIT 1

3 marks

Convert

$$4\frac{2}{3} = \frac{14}{3}, \quad 1\frac{1}{5} = \frac{6}{5}.$$

Invert and multiply

$$\frac{14}{3} \div \frac{6}{5} = \frac{14}{3} \times \frac{5}{6} = \frac{70}{18} = \frac{35}{9}.$$

Final answer

$$\frac{35}{9} = 3\frac{8}{9}$$

Question 13

4MA1-2H Q2

ORIGINAL QUESTION

3 marks

- 2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

Colour	green	yellow	brown	pink
Probability	0.32	0.13	0.28	

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

Worked Solution 13

Expected frequency from a spinner

MODULAR UNIT 1

3 marks

Find pink probability

$$P(\text{pink}) = 1 - (0.32 + 0.13 + 0.28) = 0.27.$$

Scale to 200 spins

$$0.27 \times 200 = 54.$$

Final answer

54

Question 14

4MA1-2H Q3

ORIGINAL QUESTION

4 marks

- 3 $ABCD$ is a trapezium.

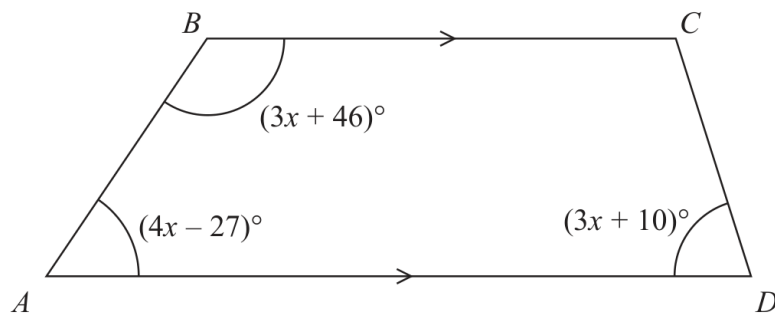


Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

°

Worked Solution 14

Largest angle in a trapezium

MODULAR UNIT 1

4 marks

Use parallel-line angles

$$(4x - 27) + (3x + 46) = 180 \Rightarrow 7x + 19 = 180 \Rightarrow x = 23.$$

Evaluate the angles

$$3x + 46 = 115, \quad 3x + 10 = 79, \quad 4x - 27 = 65; \text{ the remaining angle is } 101.$$

Final answer

$$115^\circ$$

Question 15

4MA1-2H Q4

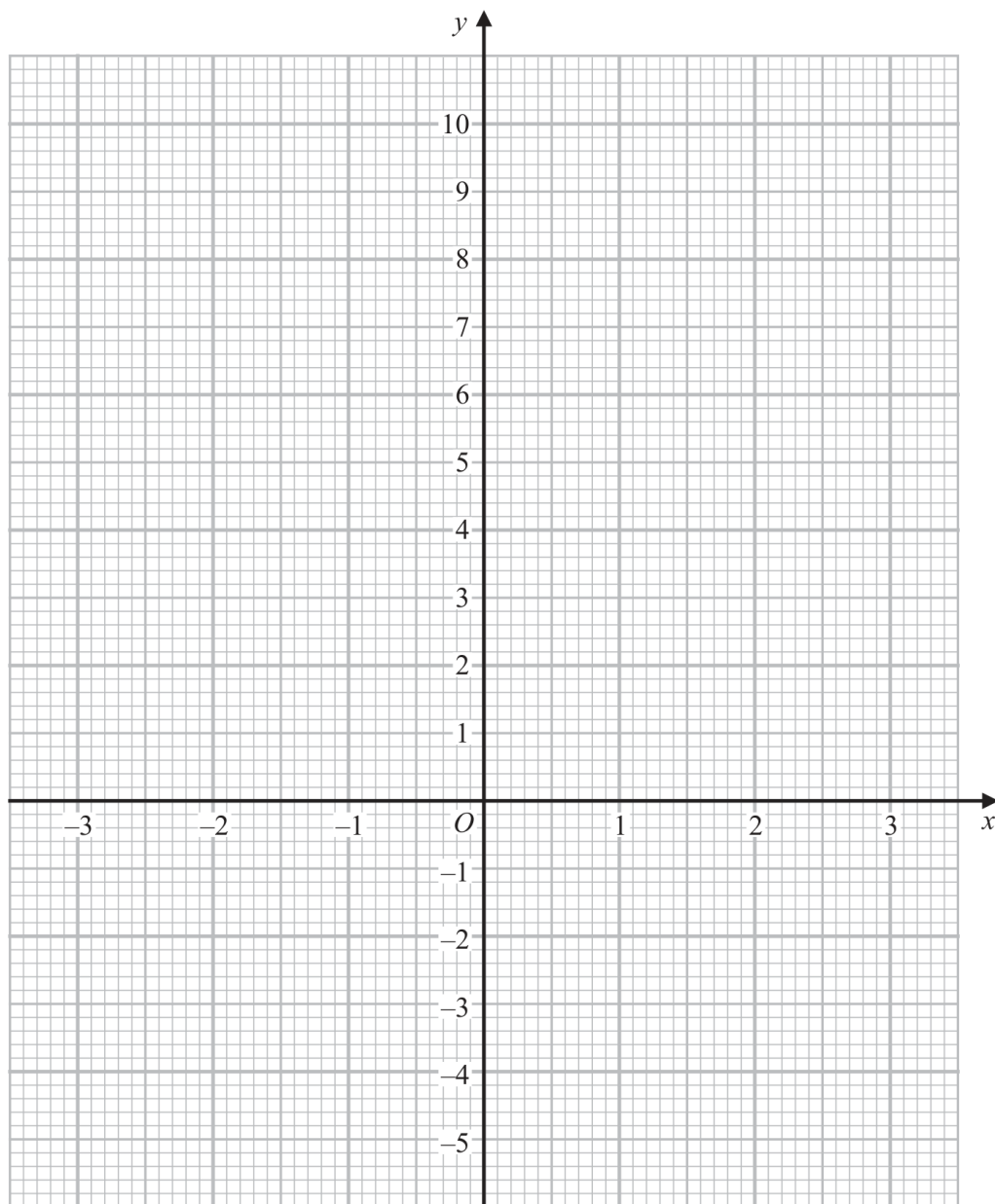
ORIGINAL QUESTION**4 marks**

- 4 (a) Complete the table of values for $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

(2)

- (b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of x from -3 to 3



(2)

Worked Solution 15

Complete and draw a quadratic graph

MODULAR UNIT 1

4 marks

Complete the table

$$y = x^2 - x - 4 \Rightarrow (8, 2, -2, -4, -4, -2, 2) \text{ for } x = -3, -2, -1, 0, 1, 2, 3.$$

Plot and join

Plot $(-3, 8), (-2, 2), (-1, -2), (0, -4), (1, -4), (2, -2), (3, 2)$ and join with a smooth quadratic curve.

Final answer

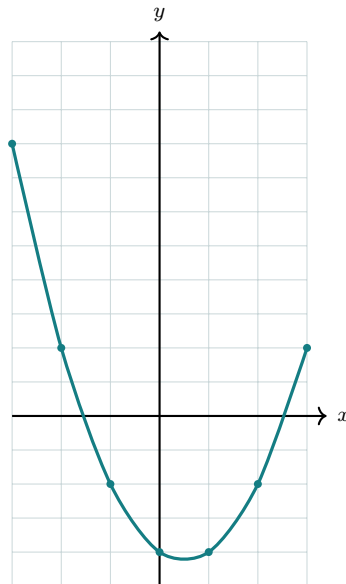
$$y = x^2 - x - 4; \quad y\text{-values } 8, 2, -2, -4, -4, -2, 2$$

Completed diagram 15

Complete and draw a quadratic graph

MODULAR UNIT 1

4 marks

Completed solution diagram: Completed quadratic graph

Question 16

4MA1-2H Q9

ORIGINAL QUESTION

3 marks

9 Change a speed of 27 kilometres per hour to a speed in metres per second.

..... m/s

Worked Solution 16

Convert kilometres per hour

MODULAR UNIT 1

3 marks

Convert distance and time

$$27 \text{ km} = 27\,000 \text{ m}, \quad 1 \text{ hour} = 3600 \text{ s}.$$

Divide

$$\frac{27\,000}{3600} = 7.5 \text{ m/s}.$$

Final answer

$$7.5 \text{ m/s}$$

Question 17

4MA1-2H Q12

ORIGINAL QUESTION

4 marks

12 The diagram shows right-angled triangle ABD

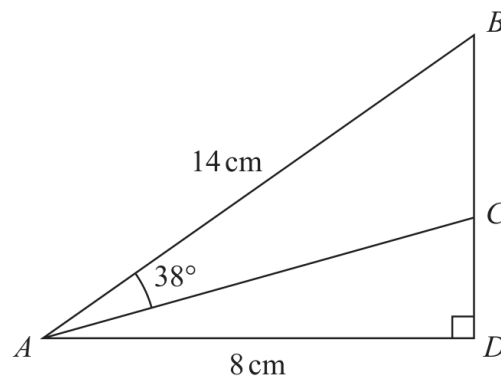


Diagram **NOT**
accurately drawn

$$AB = 14 \text{ cm} \quad AD = 8 \text{ cm}$$

C is the point on BD such that angle $BAC = 38^\circ$

Work out the length of CD

Give your answer correct to 3 significant figures.

..... cm

Worked Solution 17

Two right triangles and a split hypotenuse

MODULAR UNIT 1

4 marks

Find the angle in ABD

$$\cos(\angle BAD) = \frac{AD}{AB} = \frac{8}{14} \Rightarrow \angle BAD \approx 55.15^\circ.$$

Find the angle in ACD

$$\angle CAD = 55.15 - 38 \approx 17.15^\circ.$$

Find CD

$$\tan(17.15^\circ) = CD/8 \Rightarrow CD \approx 2.47 \text{ cm}.$$

Final answer

2.47 cm

Question 18

4MA1-2H Q13

ORIGINAL QUESTION**4 marks**

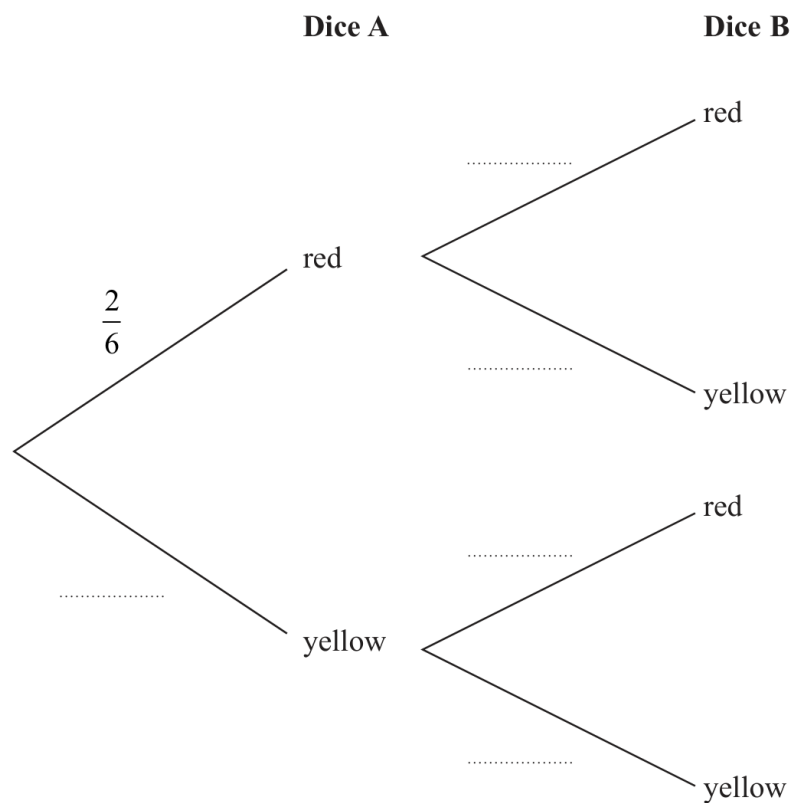
13 Narin has two fair 6-sided dice.

Dice **A** has 2 red faces and 4 yellow faces.

Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both dice land on yellow.

(2)

Worked Solution 18

Two-dice probability tree

MODULAR UNIT 1

4 marks

Complete the tree $A : R = 2/6, Y = 4/6; \quad B : R = 1/6, Y = 5/6$ on every second branch.**Both yellow**

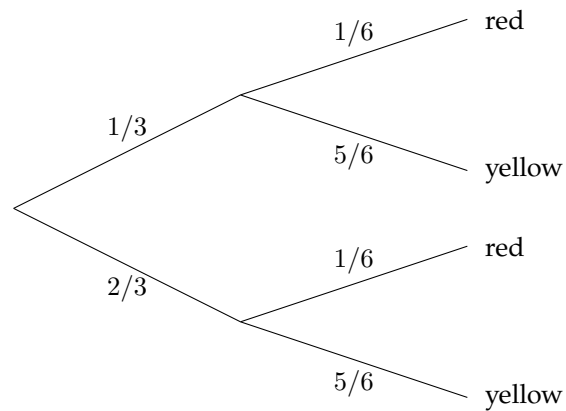
$$P(Y, Y) = \frac{4}{6} \times \frac{5}{6} = \frac{20}{36} = \frac{5}{9}.$$

Final answer

$$\frac{5}{9}$$

Completed diagram 18

Two-dice probability tree

MODULAR UNIT 1**4 marks****Completed solution diagram: Completed probability tree****Final answer**

$$P(\text{both yellow}) = \frac{2}{3} \times \frac{5}{6} = \frac{5}{9}$$

Question 19

4MA1-2H Q15

ORIGINAL QUESTION

2 marks

15 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

.....

(2)

Worked Solution 19

Fractional index power

MODULAR UNIT 1

2 marks

Apply the outer power

$$(32a^{15})^{3/5} = 32^{3/5}a^9 = (2^5)^{3/5}a^9 = 8a^9.$$

Final answer

$$8a^9$$

Question 20

4MA1-2H Q15

ORIGINAL QUESTION

2 marks

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

.....
(2)

Worked Solution 20

Negative index power

MODULAR UNIT 1

2 marks

Invert the bracket

$$\left(\frac{1}{10x}\right)^{-3} = (10x)^3.$$

Expand

$$(10x)^3 = 1000x^3.$$

Final answer

$$1000x^3$$

Question 21

4MA1-2H Q15

ORIGINAL QUESTION

3 marks

(c) Solve $\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$

Show clear algebraic working.

$y = \dots\dots\dots$
(3)

Worked Solution 21

Solve a fractional linear equation

MODULAR UNIT 1

3 marks

Clear denominators

$$30 \left(\frac{1 - 2y}{3} \right) = 30 \left(\frac{4}{5} - \frac{2y - 1}{2} \right).$$

Expand and collect

$$10 - 20y = 24 - 15(2y - 1) = 39 - 30y \Rightarrow 10y = 29.$$

Solve

$$y = \frac{29}{10} = 2.9.$$

Final answer

$$y = 2.9$$

Question 22

4MA1-2H Q17

ORIGINAL QUESTION

3 marks

17 (a) Expand and simplify $(x + 6)(3x - 2)(x + 6)$

(3)

Worked Solution 22

Expand three brackets

MODULAR UNIT 1

3 marks

Square the repeated factor

$$(x + 6)^2 = x^2 + 12x + 36.$$

Multiply by the third factor

$$(x^2 + 12x + 36)(3x - 2) = 3x^3 + 34x^2 + 84x - 72.$$

Final answer

$$3x^3 + 34x^2 + 84x - 72$$

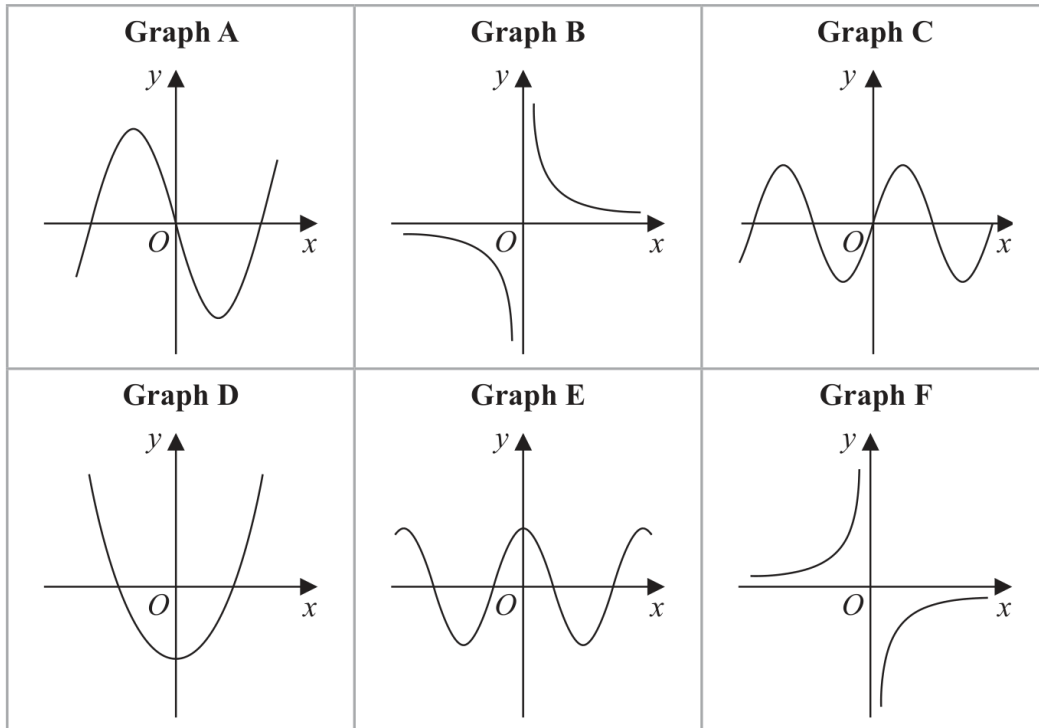
Question 23

4MA1-2H Q18

ORIGINAL QUESTION

3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	
$y = -\frac{3}{x}$	
$y = 4x^3 - 5x$	

Worked Solution 23

Recognise function graph families

MODULAR UNIT 1

3 marks

Match the sine graph $y = \sin x$ passes through the origin and oscillates : Graph C.**Match the reciprocal graph** $y = -3/x$ has positive branch for $x < 0$ and negative branch for $x > 0$: Graph F.**Match the cubic** $y = 4x^3 - 5x$ is an odd cubic with negative central gradient : Graph A.**Final answer**

$$\sin x \rightarrow C, \quad -3/x \rightarrow F, \quad 4x^3 - 5x \rightarrow A$$

Question 24

4MA1-2H Q19

ORIGINAL QUESTION

3 marks

19 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

Worked Solution 24

Complete the square

MODULAR UNIT 1

3 marks

Group the quadratic terms

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5.$$

Complete the square

$$3(x^2 - 2x) + 5 = 3[(x - 1)^2 - 1] + 5 = 3(x - 1)^2 + 2.$$

Final answer

$$3(x - 1)^2 + 2$$

Question 25

4MA1-2H Q20

ORIGINAL QUESTION

3 marks

20 There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

Worked Solution 25

At least one red counter remains

MODULAR UNIT 1

3 marks

Use the complement

The only failure is drawing all three red: $P(RRR) = \frac{3}{12} \times \frac{2}{11} \times \frac{1}{10} = \frac{1}{220}$.

Subtract from one

$$P(\text{at least one red remains}) = 1 - \frac{1}{220} = \frac{219}{220}.$$

Final answer

$$\frac{219}{220}$$

Question 26

4MA1-2H Q22

ORIGINAL QUESTION

5 marks

22 The diagram shows a triangle ABC and a flagpole BF

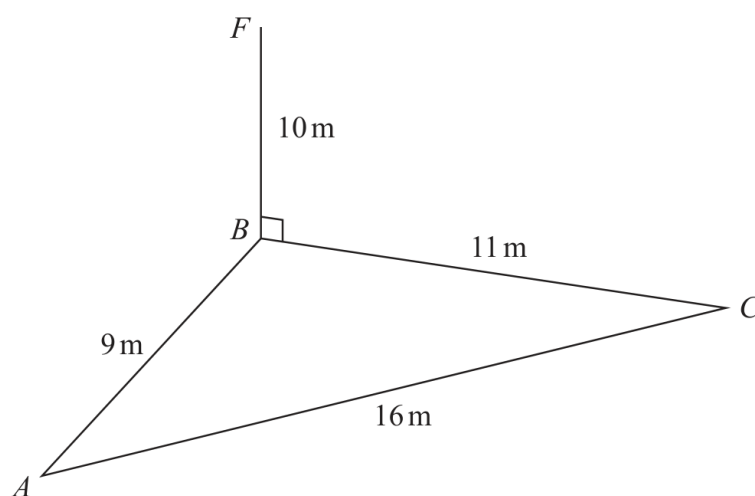


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9\text{ m} \quad BC = 11\text{ m} \quad AC = 16\text{ m} \quad BF = 10\text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D

Give your answer correct to one decimal place.

○

Worked Solution 26

Angle of elevation from a perpendicular foot

MODULAR UNIT 1

5 marks

Find the ground angle

$$\cos \angle BCA = \frac{11^2 + 16^2 - 9^2}{2(11)(16)} = \frac{37}{44} \Rightarrow \angle BCA \approx 32.763^\circ.$$

Find BD

$$BD = 11 \sin 32.763^\circ \approx 5.95 \text{ m.}$$

Use the vertical flagpole

$$\tan \angle FDB = \frac{BF}{BD} = \frac{10}{5.95}.$$

Evaluate

$$\angle FDB \approx 59.2^\circ.$$

Final answer

$$59.2^\circ$$

Question 27

4MA1-2H Q23

ORIGINAL QUESTION

4 marks

- 23 The diagram shows a cuboid with a square cross section.

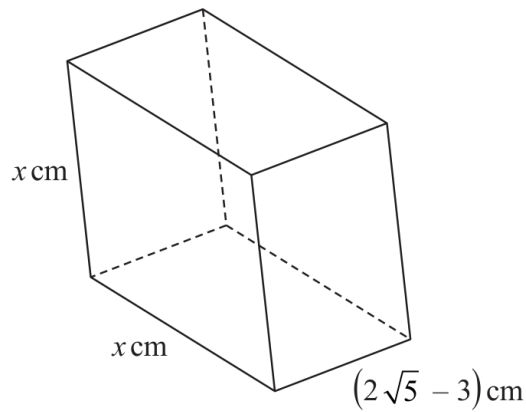


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{ cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x =$

Worked Solution 27

Surd volume and a square cross-section

MODULAR UNIT 1

4 marks

Form x squared

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5} \Rightarrow x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}.$$

Rationalise

$$x^2 = \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{11} = 9 + 4\sqrt{5}.$$

Recognise the square

$$9 + 4\sqrt{5} = (2 + \sqrt{5})^2.$$

Take the positive length

$$x = 2 + \sqrt{5}.$$

Final answer

$$2 + \sqrt{5} \text{ cm}$$

Question 28

4MA1-2H Q24

ORIGINAL QUESTION

6 marks

24 $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(.....,)

Worked Solution 28

Coordinates of the fourth vertex of a kite

MODULAR UNIT 1

6 marks

Find AC

$$m_{AC} = \frac{-5 - 10}{4 - (-2)} = -\frac{5}{2}, \quad y = 5 - \frac{5}{2}x.$$

Use the perpendicular diagonal

$$m_{BD} = \frac{2}{5}, \quad y - 4 = \frac{2}{5} \left(x + \frac{27}{5} \right).$$

Find the diagonal intersection

$$AC \cap BD = \left(-\frac{2}{5}, 6 \right).$$

Reflect B in AC

$$D = 2 \left(-\frac{2}{5}, 6 \right) - \left(-\frac{27}{5}, 4 \right) = \left(\frac{23}{5}, 8 \right).$$

Final answer

$$D = (4.6, 8)$$

Question 29

4MA1-2H Q25

ORIGINAL QUESTION

4 marks

- 25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.
- Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

..... g/cm^3

Worked Solution 29

Upper bound for sphere density

MODULAR UNIT 1

4 marks

Choose the worst-case bounds

$$r_{LB} = 2.75 \text{ cm}, \quad M_{UB} = 265\pi \text{ g}.$$

Use the lower volume

$$V_{LB} = \frac{4}{3}\pi(2.75)^3.$$

Maximise density

$$\rho_{UB} = \frac{265\pi}{(4/3)\pi(2.75)^3} = 9.56 \text{ g/cm}^3 \text{ (2 d.p.)}.$$

Final answer

$$9.56 \text{ g/cm}^3$$